



Saint Columban College
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Caesar

caesar 



Medium

Cryptography

picoCTF 2019

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Description

Decrypt this message.

Message: [message](#)

Hints 

1

Caesar cipher tutorial

57,021 users solved



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picoCTF{FLAG}

Submit Flag

Author: The Analyst: Hyposelenia

Challenge: caesar

Category: Cryptography

Date: 02/26/26



I. Objective

The objective of this challenge is to **decrypt a message encrypted using the Caesar cipher** by identifying the correct shift value and retrieving the hidden flag.

II. Background

The Caesar cipher is one of the **oldest and simplest encryption techniques**. It is a type of **substitution cipher**, where each letter in the plaintext is shifted a fixed number of positions forward in the alphabet.

For example:

- With a shift of 3:

$$A \rightarrow D, B \rightarrow E, C \rightarrow F$$

This cipher was historically used by **Julius Caesar** to send secret military messages. However, because there are only **25 possible shifts**, the Caesar cipher is extremely weak and can be broken easily using **brute force**.

Today, the Caesar cipher is mainly used for **learning purposes** to introduce the fundamentals of cryptography.

III. Tool Used

- **Caesar Cipher Online Decoder** - Used to test different shift values and reveal the original plaintext from the encrypted message.
- **Microsoft Word** - Used to open and view the provided text file.

IV. Methodology

1. Downloaded the provided file from the challenge description.



Description

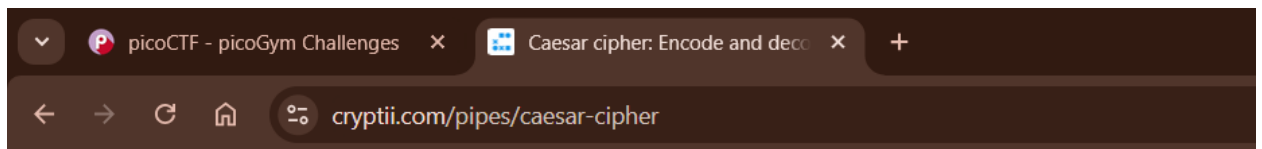
Decrypt this message.

Message: **message**

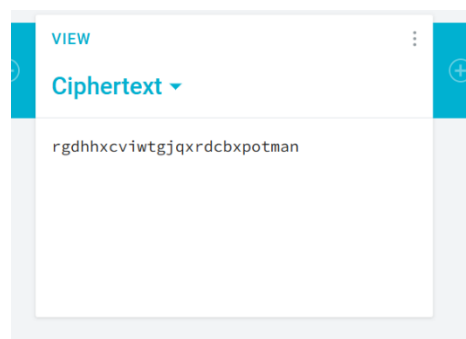
2. Opened the file using Microsoft Word to view the encrypted text.

```
picoCTF{rgdhxvcviwtgjxrdbcxpotman}
```

3. Copied the shifted letters from the file.
4. Searched for an online Caesar cipher decoder.



5. Pasted the encrypted text into the decoder.





6. Tried different shift values one by one.

SHIFT

— 15 a→p +

7. Identified that **shift value 15** produced a readable and meaningful sentence.

VIEW Ciphertext ▾

rgdhxvcviwtgjxrdcbxpotman

ENCODE DECODE

Caesar cipher ▾

SHIFT — 15 a→p +

ALPHABET

abcdefghijklmnopqrstuvwxyz

CASE STRATEGY FOREIGN CHARS

Maintain case ▾ Include Ignore

→ Decoded 28 chars

VIEW Plaintext ▾

crossingtherubiconmiazexly

8. Extracted the decoded text.

9. Placed the decoded text inside the required **picoCTF{}** format.

V. Result

picoCTF{crossingtherubiconmiazexly}

VI. Explanation

What is the Caesar Cipher?

The Caesar cipher shifts letters in the alphabet by a fixed number. When the alphabet reaches Z, it **wraps around** back to A.

Example with shift 2:

- $Y \rightarrow A$
- $Z \rightarrow B$

Why Was This Easy to Break?



There are only **25 possible shifts**. This means an attacker can simply:

- Try every shift
- Look for readable English text
- Stop once the message makes sense

This process is called a **brute-force attack**.

How Did We Know the Correct Shift?

At shift **15**, the decoded message:

- Formed real English words
- Had a meaningful sentence structure
- Looked like a typical picoCTF flag format

Because only one shift produces readable English, identifying the correct one is straightforward.

Why Is Caesar Cipher Insecure?

- No keys
- No randomness
- Very small keyspace
- Vulnerable to brute force and frequency analysis

This makes it unsuitable for real-world security.

VII. Conclusion

The Caesar challenge demonstrates how **simple encryption methods can be easily broken** using brute-force techniques. By systematically testing all possible shifts, the original message was successfully decrypted.

This challenge reinforces the importance of using **modern cryptographic algorithms** instead of classical ciphers when protecting sensitive data.